

RAYANE DAKHLAOU

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RESEARCH INTERESTS

Representation & Self-Supervised Learning · 3D Computer Vision & Geometric Deep Learning · World Models & Physical AI

RESEARCH EXPERIENCE

AI Research Intern — Siemens AG

Apr. 2026 – Sept. 2026

Dept. Foundational Technologies, Digital Manufacturing Technologies Research Group

Munich, Germany

- Designing a novel Self-Supervised Representation Learning framework for Boundary-Representation (3D CAD models) built on a Joint-Embedding Predictive Architecture (JEPA).
- Built a geometry encoder over a geometric Attributed Adjacency Graph, coupling Cross-Scale Mutual Attention with edge-aware Message Passing (Graph Transformer); benchmarked enhanced Point Transformer v3 point-cloud backbone (Utonia) as alternative encoder.
- Ran an instrumented study isolating what latent-prediction objective *can and cannot* force; frozen linear-probe transfer to 0.71 accuracy (+0.38 over a matched random-init control) in the label-scarce regime (<0.5% labels).
- Engineered a geometry-driven module auto-generating PMI (GD&T tolerances & datums) from CAD geometry to automate CMM inspection planning.
- Designed the pretraining for few-shot transfer across downstream tasks — automated machining-feature recognition (CAM), part segmentation & classification.

EDUCATION

PSL Research University — ENS Paris & Mines Paris & Paris-Dauphine

Sept. 2025 – Dec. 2026

MSc IASD (Artificial Intelligence, Systems & Data) — Mathematics Track

- Highly selective research-oriented master in Applied Mathematics, Learning & Modeling.
- Key courses :** Reinforcement Learning · Advanced Machine Learning · Convex Optimization · Dimension Reduction and Manifold Learning · High-Dimensional Statistics · Bayesian Statistics · Optimal Transport for ML · Machine Learning with Kernel Methods · Large Language Models for Code & Proof · 3D Point Cloud and Modeling · Deep Learning for Computer Vision

Télécom Paris — Institut Polytechnique de Paris

Sept. 2023 – Dec. 2026

Engineering Degree (M.Sc.), Applied Mathematics & Computer Science

- Top-tier French Engineering school, Leader in the field of digital technology
- Stochastic Processes and Scientific Computing:** Martingales · Markov chains · Stochastic Processes (Brownian motion, Poisson process) · High-Dimensional Probability · Statistical Learning · Numerical analysis (Newton, Monte-Carlo)
- Signal Processing & Artificial Intelligence Track:** Statistics · Optimization · Time-Series Analysis · Machine Learning · Deep Learning (CNN, RNN, VAE, GAN, Transformers) · LLM · Audio & Speech processing · Databases.

Preparatory Class for French Grandes Écoles — Lycée Condorcet, Paris IX

Sept. 2021 – June 2023

Scientific Track PSI* (eq. BSc. in Mathematics and Physics)

- 2-year intensive program in Mathematics & Physics; ranked top 8% nationally in the highly competitive Grandes Écoles entrance examinations.

SELECTED PROJECTS

Kernel Methods for Image Classification | [GitHub](#) | [Kaggle](#)

- Built a full CIFAR-10 classifier using only kernel methods, implemented from scratch (custom feature kernels, CKN, KRR), with no deep-learning libraries — ranked 3rd out of 44 teams on the Kaggle Data Challenge leaderboard.

GAN Latent-Space Tuning | [GitHub](#)

- Studied the precision-recall trade-off of f-GANs across JS/KL/RKL/BCE objectives; implemented Soft Truncation & Discriminator Rejection Sampling to improve post-hoc sample quality.

Collaborative Filtering | [GitHub](#)

- Developed a Multiple-Kernel Matrix Factorization recommender system on scikit-learn, accelerated with Numba.

hERG Cardiotoxicity Prediction — Hackathon IBM × Qubit Pharmaceuticals × Margo | [GitHub](#)

- Finalist (Top 4/25 teams) in a 56-hour hackathon on in-silico hERG-blockade prediction from molecular structure; built an ensemble of GNN, XGBoost & ChemBERTa for toxicity prediction and Tanimoto-distance clustering.

DeepSDF In-Depth Study — Implicit Neural 3D Shape Representation

- Reimplemented DeepSDF to learn continuous signed-distance fields as implicit neural representations of 3D geometry; used latent-code optimization for shape reconstruction & completion from partial observations.

2PAC — Asynchronous BFT Blockchain Consensus | [Paper](#) | [GitHub](#)

- Implemented & benchmarked the 2PAC consensus protocol (with Prof. M. Rambaud); stress-tested under Byzantine node corruption & network latency via socket-level simulation.

TECHNICAL SKILLS

Programming: Python · C · SQL · TypeScript · $\text{\LaTeX}$

ML & scientific stack: PyTorch · NumPy · Pandas · scikit-learn · TensorFlow/Keras · Numba · Git

Languages: French (native) · English (full professional) · Arabic (intermediate) · German (beginner)